

NADIA ISLAM

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EDUCATION

City University of New York, Queens College

Queens, NY

Bachelor of Science in Computer Science | GPA: 3.0 / 4.0

Expected Dec 2026

- **Relevant Coursework:** Object-Oriented Programming (Java & C++), Data Structures, Algorithms, Discrete Mathematics, Discrete Structures
- **Honors & Programs:** Google x Bastas Software Engineering Program (G-SWEP) – Fall 2024; Google CS Summer Institute (CSSI) – 2022
- **Activities:** Girls Who Code – Mentor Corps mentor (20+ K–12 students) & Queens College Chapter member, Sep 2022–May 2025

TECHNICAL SKILLS

- **Languages:** Java, C++, Python, JavaScript, SQL, HTML, CSS
- **Frameworks & APIs:** PySpark, Flask, Node.js, React, Discord.js, OpenAI API, Google Sheets API, REST APIs, JSON
- **Cloud & DevOps:** AWS (EC2, S3, Lambda, VPC, Security Groups), Docker, Kubernetes (Minikube, kubectl), Git, GitHub, Linux/Unix CLI
- **Distributed Systems & Data:** Hadoop (HDFS, YARN), Apache Spark, Redis Cluster, distributed storage, replication, failover, sharding
- **Concepts:** Object-Oriented Programming, Data Structures & Algorithms, API Integration, Automation, Performance Benchmarking, HIPAA-aware data handling

TECHNICAL PROJECTS

Distributed Redis Cluster with Failover Testing on AWS EC2 (Python, Redis, AWS)

Apr 2026

- Architected and deployed a 6-node Redis Cluster (3 masters, 3 replicas) across AWS EC2 with 16,384 hash slots, configuring VPC security groups, SSH key pairs, and gossip-protocol coordination.
- Built a multithreaded Python workload generator (8–10 threads, mixed GET/SET/INCR/DEL) and benchmarked the cluster at 4,023 ops/sec with 2.47 ms avg latency, P95 6.22 ms, P99 9.86 ms over 241K operations.
- Performed live failover testing by killing a master mid-workload; documented automatic replica promotion, slot reassignment, and quantified throughput degradation. Repo: github.com/isla6645/redis-cluster-project

Hadoop & Spark Cluster on AWS EC2 (Hadoop, Spark, PySpark, AWS)

Apr 2026

- Provisioned a 3-node Hadoop/Spark cluster on AWS EC2 (1 master, 2 workers) with passwordless SSH, configured HDFS NameNode and DataNodes, and validated cluster health via hdfs fsck and the NameNode Web UI.
- Wrote PySpark applications for top-20 word-frequency analysis over Project Gutenberg text and a Monte Carlo π estimation across 6 partitions, executed on the Spark master/worker cluster.

Containerized Flask App with Docker & Kubernetes (Docker, Kubernetes, Minikube, Python)

Mar 2026

- Containerized a Python Flask service with Docker and deployed it to a local Minikube Kubernetes cluster using YAML manifests for Deployment and NodePort Service resources.
- Demonstrated horizontal scaling (2→4 replicas), zero-downtime rolling updates (v1→v2 via kubectl set image), and load balancing across pods verified by hostname rotation on repeated requests.

PROFESSIONAL EXPERIENCE

Shah's Halal Food – Operations & Technical Support Associate

Queens Village, NY

Apr 2025 – Present

- Built Python automation integrated with the Google Sheets API to reconcile inventory and aggregate daily sales across multiple locations, cutting manual data-entry time by 40%.
- Configured and troubleshoot POS systems and third-party delivery integrations (Uber Eats, Grubhub, DoorDash) by consuming RESTful APIs, sustaining 99%+ platform uptime.
- Deployed custom alerting and routing workflows for incoming orders, reducing average order-processing latency by 15% and improving delivery throughput.

Star Care Pharmacy – Pharmacy Technology & Operations Assistant

Jamaica, NY

Jan 2025 – Present

- Supported backend operations on the BestRx pharmacy management platform with exposure to HIPAA-compliant systems, secure patient-data workflows, and healthcare data integrity practices.
- Prototyped an LLM-powered medication FAQ assistant (OpenAI API) translating prescription details, dosing, and side-effects into plain-language responses for non-technical customers.
- Identified workflow bottlenecks with pharmacists and proposed AI-driven automation for repetitive admin tasks such as insurance prior-authorization follow-ups.